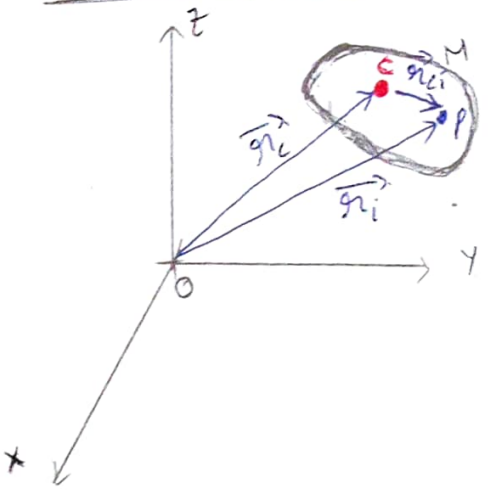
Center of mass



→ A body of mass M is placed in a frame of reference.

→ If this body is a collection of N particles such that mass of ith particle is m_i and its position is $\vec{r}_i$.

→ Now consider a point C in the system whose position vector is $\vec{r}_c$. In $\triangle^{le}$ OCP:

$$\vec{r}_i = \vec{r}_c + \vec{r}_{ci}$$

Now multiply with the mass m_i of the particle

$$\therefore \quad m_i \vec{r}_i = m_i \vec{r}_c + m_i \vec{r}_{ci}$$

Sum over all particles.

$$\sum_N m_i \vec{r}_i = \sum_N m_i \vec{r}_c + \sum_N m_i \vec{r}_{ci}$$

for "C" to be center of mass, the masses should be distributed such that

$$\sum_N m_i \vec{r}_{ci} = 0$$

i.e; mass of the system is distributed equally about C.

Then $$\sum_N m_i \vec{r}_i = \sum_N m_i \vec{r}_c$$

$$\therefore \vec{r}_c = \frac{\sum_N m_i \vec{r}_i}{\sum_N m_i}$$

$\sum_N m_i = M$ — Total Mass of the body

$$\therefore \boxed{\vec{r}_c = \frac{1}{M} \sum_N m_i \vec{r}_i}$$ position vector of center of mass.

$$= \frac{1}{M} (m_1 \vec{r}_1 + m_2 \vec{r}_2 + \cdots m_N \vec{r}_N)$$

In terms of x, y, z coordinates

$$\vec{r}_c = x_c \hat{i} + y_c \hat{j} + z_c \hat{k}$$

$$x_c = \frac{1}{M} (m_1 x_1 + m_2 x_2 + \cdots m_N x_N)$$

$$y_c = \frac{1}{M} (m_1 y_1 + m_2 y_2 + \cdots m_N y_N)$$

$$z_c = \frac{1}{M} (m_1 z_1 + m_2 z_2 + \cdots m_N z_N)$$

→ **Motion of Center of mass:**

$$\vec{r}_c = \frac{1}{M} \sum_N m_i \vec{r}_i$$

$$\frac{d\vec{r}_c}{dt} = \frac{1}{M} \frac{d}{dt} \sum_N m_i \vec{r}_i$$

$$= \frac{1}{M} \sum_N m_i \frac{d\vec{r}_i}{dt}$$

$$\therefore \boxed{\vec{v}_c = \frac{1}{N} \sum_N m_i \vec{v}_i}$$ velocity of the center of mass

$$\vec{v}_c = \frac{1}{M}\sum_N m_i \vec{v}_i$$

$$= \frac{1}{M}\sum_N \vec{p}_i$$

$$\boxed{M\vec{v}_c = \vec{p}_{net}}$$

Total momentum of the system is equal to the product of total mass M & the velocity of COM of the system.

$$\vec{v}_c = \frac{1}{M}\sum_N m_i \vec{v}_i$$

$$\frac{d\vec{v}_c}{dt} = \frac{1}{M}\sum_N m_i \frac{d\vec{v}_i}{dt}$$

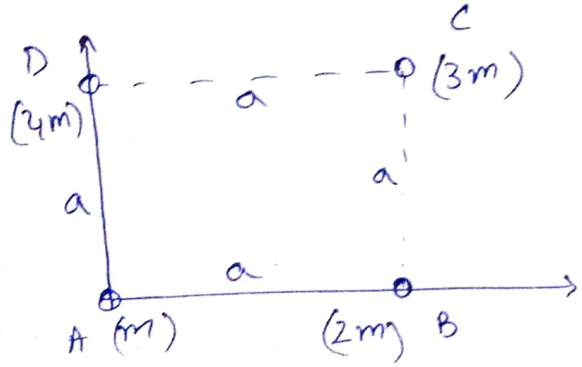
$$\vec{a}_c = \frac{1}{M}\sum_N m_i \vec{a}_i$$

$$\vec{a}_i = \frac{\vec{F}}{M}$$

$$\vec{F} = M\vec{a}_c$$

} Total force acting on the system is proportional to the acceleration of Center of Mass.

10)



Find the C.O.M of such a system.

Coordinates of ABCD are

	X	Y
A	0	0
B	a	0
C	a	a
D	0	a

$$X_{CM} = \frac{1}{M}\left(m_1x_1 + m_2x_2 + m_3x_3 + m_4x_4\right)$$

$$= \frac{1}{(4m+3m+2m+m)}\left(0 + 2ma + 3ma + 0\right)$$

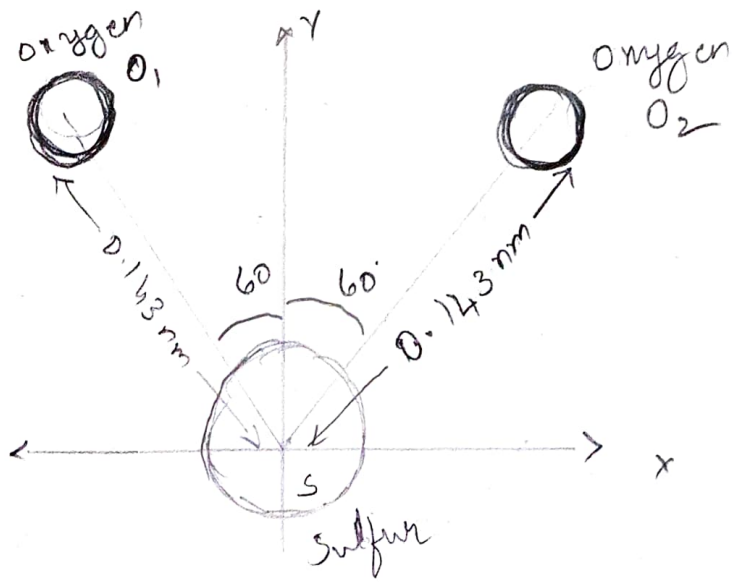
$$= \frac{5ma}{10m} = \frac{a}{2}$$

$$Y_{CM} = \frac{1}{M}\left(m_1y_1 + m_2y_2 + m_3y_3 + m_4y_4\right)$$

$$= \frac{1}{10m}\left(0 + 0 + 3ma + 4ma\right)$$

$$= \frac{7ma}{10m} = \frac{7}{10}a$$

2(a)

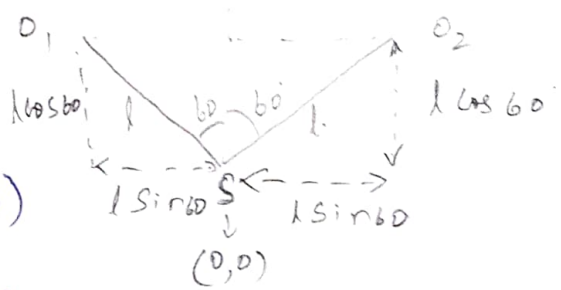


→ The drawing shows a SO_2 molecule. It consists of 2 oxygen atoms and a sulphur atom. Sulfur atom is twice as massive as oxygen atom. Using this information and the drawing, find the C.O.M of SO_2 molecule.

Sol

$m_s = 2m_o$ (mass of sulfur is 2 times mass of oxygen)

	x	y
S -	0	0



$O_1 - (-l\sin 60, +l\cos 60)$

$O_2 - (l\sin 60, l\cos 60)$

$$x_{cm} = \frac{1}{M}(m_1x_1 + m_2x_2 + m_3x_3) \quad (M = m_1 + m_2 + m_3)$$

$$x_{cm} = \frac{1}{(m_s + 2m_o)}\left(m_o(-l\sin 60) + m_s \cdot 0 + m_o(l\sin 60)\right)$$

$$x_{cm} = \frac{1}{4m_o}(0) = \underline{0}$$

$$y_{cm} = \frac{1}{M}(m_1y_1 + m_2y_2 + m_3y_3)$$

$$= \frac{1}{4m_o}(m_o l\cos 60 + m_s \cdot 0 + m_o l\cos 60)$$

$$= \frac{2m_o(l\cos 60)}{4m_o} = \frac{1}{2}\cos 60 = \frac{1}{4}$$

$$y_{cm} = \frac{0.243\,nm}{4} = \underline{0.036\,nm}$$